IRZ

$G = (V, T, P, S)$

- T → TERMINALI
- S → ZAČETNI SIMBOL
- V → SPREMENLJIVKE → NEKONČNE STVARI
- P → PRODUKCIJE $A \to \alpha$

PRIMER:

$S \to ABA$

$A \to Aa \mid a$

$B \to Bbb \mid \varepsilon$

→ P (PRAVILA)

$\begin{cases} A \to a \\ A \to Aa \end{cases}$

$G = (\{S, A, B\}, \{a, b\}, P, S)$

KAJ NARDIMO?

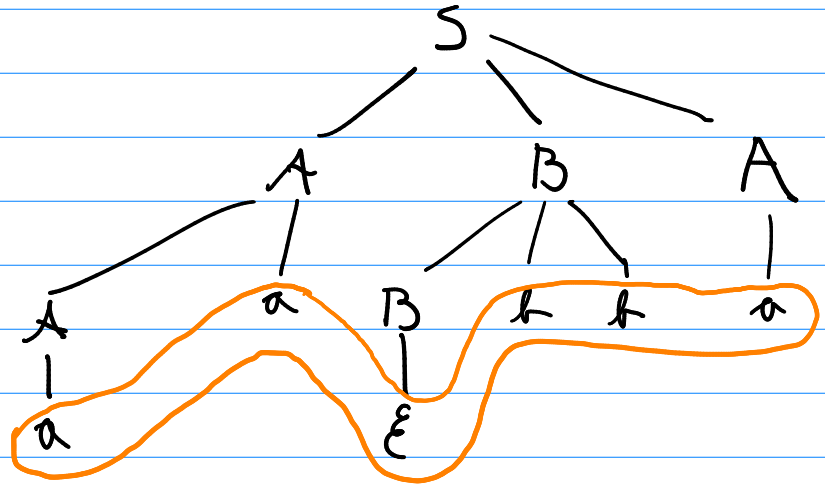
$w = aabba$

$S \Rightarrow ABA \Rightarrow AaBA \Rightarrow aaBa \Rightarrow aaBa \Rightarrow aaBbba \Rightarrow \boxed{aabba}$

$L(G) \to \{ w \in T^*, S \overset{*}{\Rightarrow} w \}$

→ JEZIK

DREVO IZPELJAVE



PALINDROMI

$S \to \varepsilon \mid a \mid b \mid aSa \mid bSb$

OBSTOJEČ PALINDROM

1) GNEZDENI OKLEPAJI

$S \Rightarrow \varepsilon \mid (SS)$

2) $L = \{0^i 1^j 0^k \mid i+j \leq 2k\}$

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$S \to AB$

$A \to \varepsilon \mid 0A$

$B \to \varepsilon \mid 1B \mid 110B$

3)

$A \Rightarrow$ POLJUBNO MNOGO "a" jev
$C \Rightarrow$ —||— "c"

$S_1 \overset{*}{\Rightarrow} a^i b^j \ (i > j)$
$S_2 \overset{*}{\Rightarrow} a^i b^j \ (i < j)$
$S_3 \overset{*}{\Rightarrow} b^j c^k \ (j > k)$
$S_4 \overset{*}{\Rightarrow} b^j c^k \ (j < k$

$S \Rightarrow S_1C \mid S_2C \mid AS_3C \mid AS_4$

$A \Rightarrow \varepsilon \mid Aa$
$C \Rightarrow \varepsilon \mid Cc$
$S_1 \Rightarrow a \mid aS_1 \mid aS_1b$

$S_2 \Rightarrow b \mid S_2 b \mid a S_2 b$

$S_3 \Rightarrow b \mid b S_3 \mid b S_3 c$

$S_4 \Rightarrow c \mid S_4 c \mid b S_4 c$

4)

$R = \emptyset$	$S \Rightarrow \varepsilon$
$R = \varepsilon$	$S \Rightarrow a$
$R = a$	$S_1 \Rightarrow L(R_1)$
$R = R_1 + R_2$	$S_2 \Rightarrow L(R_2)$
$R = R_1 R_2$	$S \Rightarrow S_1 \mid S_2$
$R = R_1^*$	$S = S_1 S_2$
	$S = \varepsilon \mid S S_1$

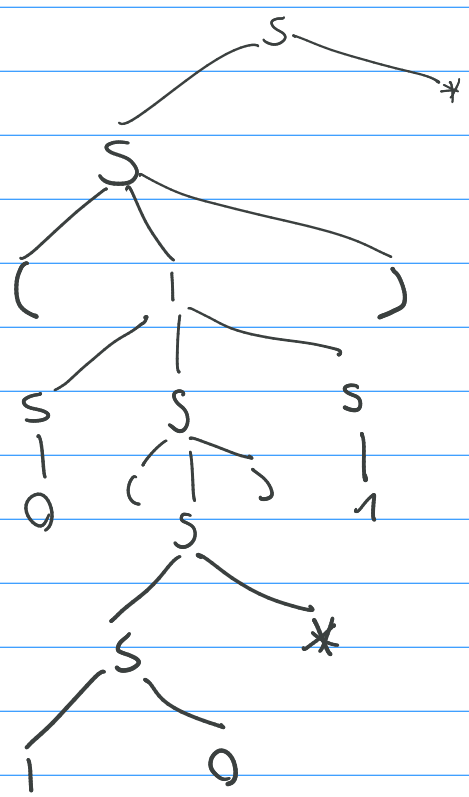
SIMBOL V JEZIKU

6)

$S \Rightarrow \emptyset \mid e \mid 0 \mid 1$

$S \Rightarrow S+S \mid SS \mid S^* \mid (S)$

$(0 + (10)^* 1)^*$



3p "0" in 5g "1" ALI JE REGULAREN?

$S \to A_{00}$

$A_{00} \to 0A_{10} \mid 1A_{01}$

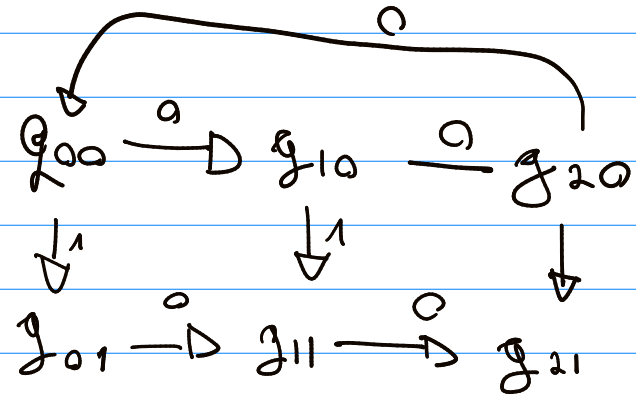
$A_{10} \to 0A_{20} \mid 1A_{11}$

$A_{20} \to 0A_{00} \mid 1A_{21}$

$A_{01} \to 0A_{11} \mid 1A_{02}$

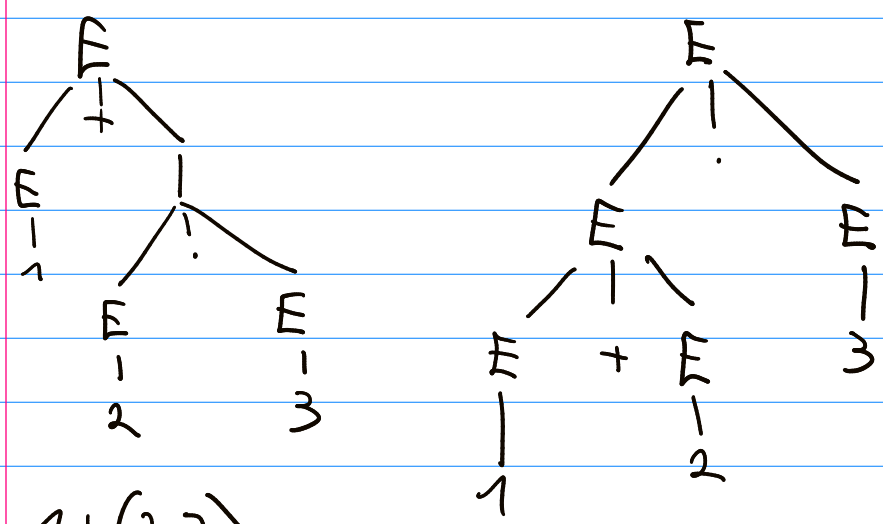
$A_{11} \to$

$A_{21} \to \ldots$



ALI JE GRAMATIKA DVOUMNA?

$E \to E+E \mid E \cdot E \mid 0 \mid 1 \sim 9 \mid$



$1+(2 \cdot 3)$

$E \to B \mid E+B$
$B \to C \mid B \cdot C$
$C \to 0-9$

- JE NEDVOUMNA ✓
- · IMA PRIORITETO NAD + ✓

$E \to B \mid E \cdot B$ → 1. NIVO, NIŽJA PRIORITETA
$B \to C \mid B+C$
$C \to 0-9$

PASSAŠ PO NIVOJIH
DOBR TRICK...